

FS Organizer

User manual

Version 1.2.0

Distributed under the GNU General Public License version 2.

The typeface is Archivo, by Omnibus-Type, under the SIL Open Font License 1.1.

Contents

1 What this program does **2**
- 1.1 What the program does not store . . . 2
- 1.2 Profile . . . 2
- 1.3 Link type . . . 2

2 Presets **3**
- 2.1 What a preset holds . . . 3
- 2.2 A preset that matches your setup . . . 3
- 2.3 The three apply modes . . . 3
- 2.4 What Replace disables by omission . . . 4
- 2.5 The return preset . . . 4
- 2.6 Presets and startup entries . . . 5

3 First setup **6**

4 Library **7**
- 4.1 The columns . . . 7
- 4.2 Enabling and disabling in bulk . . . 8
- 4.3 Categories . . . 8
- 4.4 Pinning the destination . . . 8
- 4.5 Relinking . . . 8

5 Destinations **9**
- 5.1 Importing another program's folder . . . 10

6 Importing **11**
- 6.1 Verify after copying . . . 11
- 6.2 Unfinished imports . . . 12

7 Installing from an archive or a folder **13**
- 7.1 What is refused . . . 13
- 7.2 What each Windows opens . . . 14
- 7.3 If an installation is interrupted . . . 14

8 Quarantine **15**

9 Deleting an addon from the library **16**

10 Simulator **17**
- 10.1 Startup entries . . . 17
- 10.2 Package list . . . 18

11 Documents **19**

12 Diagnostics **20**
- 12.1 Finding the culprit . . . 21

13 Journal **22**

14 Options **23**

15 Where things are kept **24**

1 What this program does

FS Organizer enables and disables Microsoft Flight Simulator addons without moving their files.

You keep your addons in a folder outside the simulator, which the program calls a **library**. Its subfolders are **categories**. Enabling an addon creates a link to it in a folder the simulator reads, called a **destination**. Disabling it removes the link. The files stay in the library the whole time, so enabling or disabling even a large addon takes no time at all.

1.1 What the program does not store

The program never writes down which addons are enabled. Every time it scans, it looks at the destinations on disk and reports what is there. If you change folders by hand, or another program's installer does, click **Refresh** and the list matches what the simulator sees.

1.2 Profile

A **profile** is one simulator installation with its destinations and libraries. If you have Flight Simulator 2020 and 2024 on the same computer, you have two profiles. The program only changes the profile in use, which you choose in **Options**.

1.3 Link type

In **Options**, on the **Links** tab, you choose how links are made.

Directory junction
Works across local drives without administrator rights. This is the default, and the one to use unless your library is on a network share.

Symbolic link
Only needed for a library on a network share. Windows only allows it with Developer Mode turned on or with administrator rights, and the program tells you when neither is available.

Changing the type affects new links only. Existing links stay as they are.

2 Presets

A preset is a named set of enabled addons. The same preset can be applied in three ways, and each gives a different result, so read the plan before you apply.

2.1 What a preset holds

To make a preset, enable the addons you want and click **New from enabled addons**.

The **Content** column shows how many addons the preset names and how many categories they are in. The **Updated** column shows when it was last saved.

A preset can also control the simulator's startup entries (section 2.6).

2.2 A preset that matches your setup

The **If applied** column shows how many addons would change if you applied the preset now, in the mode chosen under **Apply as**.

Matches your setup appears next to a preset when the enabled addons are exactly the ones it names. This check always works as Replace: if one more addon is enabled that the preset does not name, the preset no longer matches, because Replace would disable that addon. A preset you have just saved matches straight away, even though you never applied it.

Because the count follows the chosen mode and the mark does not, a preset can match your setup and still show changes. In Disable mode, for example, it would disable every addon it names.

2.3 The three apply modes

Replace
Enables the preset's addons and disables everything else.

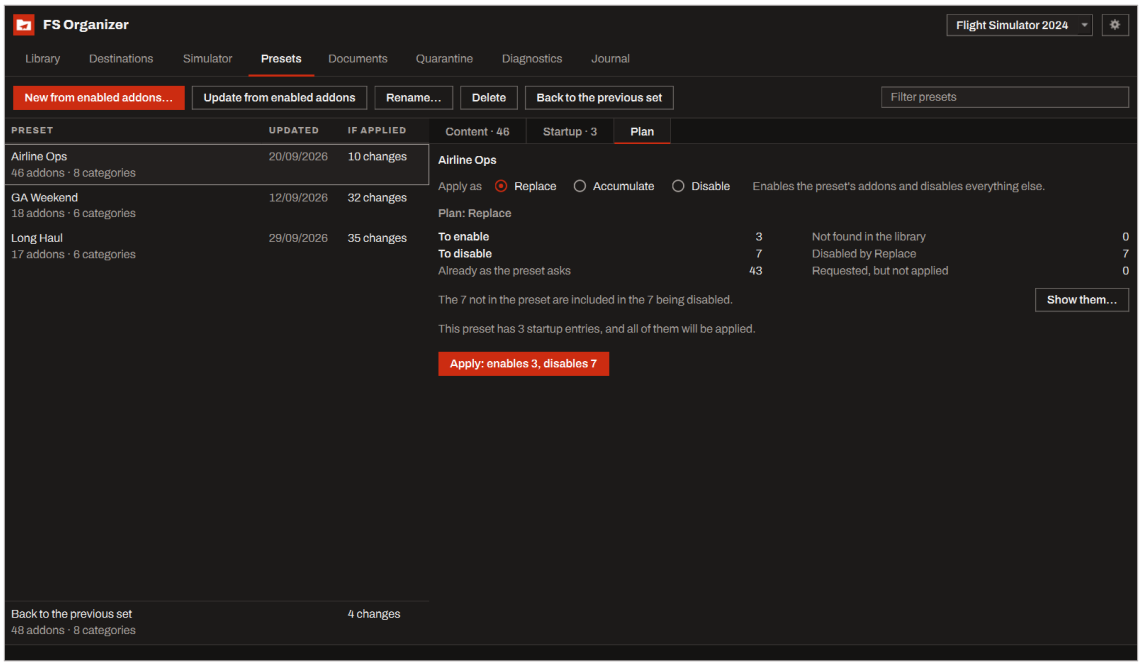
Accumulate
Enables the preset's addons and leaves the rest as it is.

Disable
Disables the preset's addons. Entries that the preset marks as disabled are ignored in this mode.

Before you apply, the **Plan** panel splits the result into five groups:

- **To enable**: addons in the preset that are disabled now.
- **To disable**: addons the preset disables, or that Replace disables because the preset does not name them.
- **Already as the preset asks**: addons that stay as they are.
- **Not found in the library**: addons the preset names that are no longer in the library. The rest of the preset is still applied.

- **Requested, but not applied**: startup entries the preset asks for that could not be changed.



The plan of the “Voo curto” preset in Replace mode. The sentence under the groups says the 4 addons outside the preset are part of the 4 being disabled.

2.4 What Replace disables by omission

In Replace mode, addons that are enabled now but not named in the preset are disabled. They are listed under **Disabled by Replace**, and **Show them** opens the list.

They are already part of the plan’s count. If the plan says it disables 12 and the list has 5 names, 12 addons are disabled in total: those 5, plus 7 the preset itself disables.

2.5 The return preset

Just before applying a preset, the program saves the addons enabled at that moment as the profile’s return preset. **Back to the previous set** applies it. If the return preset cannot be saved, for example because the presets folder is read-only, nothing is applied.

This is different from **Undo last change**:

Undo last change
Undoes what you just applied. It stops being available once you import an addon, work in the quarantine or change a category.

Back to the previous set
Applies the return preset like any other preset. It stays available after imports and quarantine work.

2.6 Presets and startup entries

On the preset's **Startup** tab, a checkbox lets the preset also control the programs that start along with the simulator. Checking it saves the startup entries that are enabled at that moment. You can then enable or disable each one there.

If startup management is off in **Options**, the preset's startup entries are not applied, and the program tells you how many were skipped. The addons are applied as usual.

If an entry the preset names is no longer in the simulator's file, its row gets the **not in the file** tag, and applying the preset leaves it out. **Remove from preset** takes the selected row out.

Add entry adds to the preset an entry of the file that it does not name yet, and leaves the other rows alone. **Update from enabled entries** does something else: it replaces the whole list with the entries enabled at that moment.

Applying a preset never adds entries to the simulator's file.

3 First setup

The first time you open the program, it looks for simulator installations and asks for two things:

The simulator
Pick the installation for this profile from the list, or choose a folder yourself. If the folder does not look like the simulator's `Community` folder, the program says so and uses it anyway.

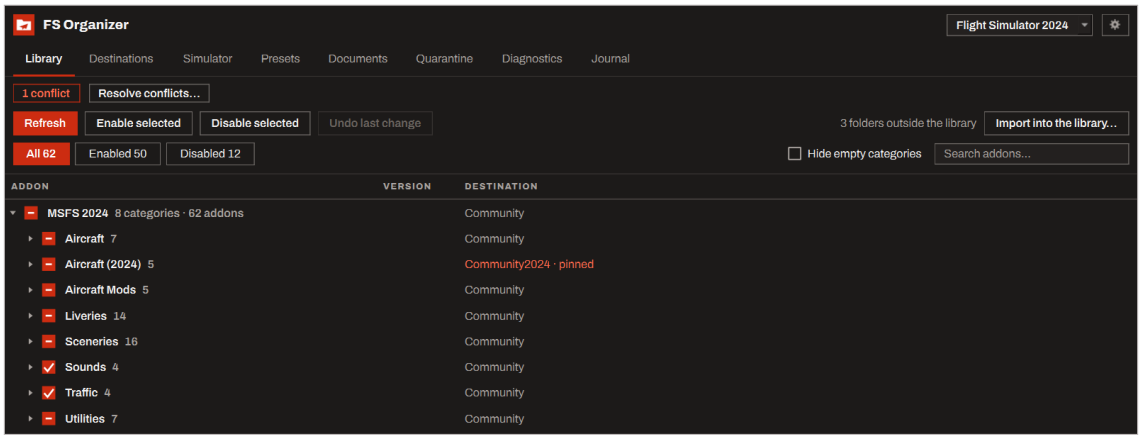
The library
Choose the folder outside the simulator where you keep your addons. Its subfolders become categories.

When you add a library with a long path, the program shows how many characters the path already uses and compares it with a short path. Windows refuses some operations, including the Recycle Bin, on paths longer than 260 characters. The program does not block you: it warns when a path gets in the way, and **Diagnostics** shows the longest path in each library.

If you used MSFS Addons Linker before, open **Options** and use **Import from MSFS Addons Linker**. It adds the libraries, categories and presets that are not here yet. No files are moved or deleted.

4 Library

The **Library** tab shows your addons as a tree, grouped by category. The checkbox on each row enables or disables the addon.



The library, with the categories open and the selected addon's panel on the right.

Three buttons above the tree filter it by state: **All**, **Enabled** and **Disabled**, each with its count. The counts do not change while you search. The search box and **Hide empty categories** work together with the button you chose, and a category shows only if it holds an addon that passes all of them.

The filter only hides rows. When you select a whole category, **Enable selected**, **Disable selected**, the category's checkbox and the other actions reach only the addons you can see in it. While a filter or a search is on, the number next to each category reads as shown out of total, for example **3 of 31**.

4.1 The columns

Addon
The folder name.

Version
The version declared in the addon's `manifest.json`.

Enabled
Whether the addon has a link in a destination right now.

Destination
Where the link is, and whether the destination is pinned.

Link
Whether the link works. If its folder is missing, the link is broken.

A row can carry one of three marks: **Broken link**, **In conflict** or **Two copies**. The sections on destinations and quarantine explain them.

4.2 Enabling and disabling in bulk

Select several rows and use **Enable selected** or **Disable selected**. For more than 10 addons, the program asks first. **Undo last change** undoes the whole batch.

If another addon of yours already uses the folder name in the destination, the program offers to swap them: it disables that one and enables yours in a single step, which you can undo in one go.

4.3 Categories

Right-click the tree to create, rename or delete a category, or to move addons between categories. Only an empty category can be deleted.

Suggest categories looks at folder names and `manifest.json` files and proposes where to move each addon. The reliable rules come checked. The livery rule is often wrong, so it comes unchecked: review it before applying.

The program puts a hidden `.fsorg-category` file in each category folder it knows about, so an empty category is still recognized as one. It marks them when you add the library, and **Categories** in the options can mark them again or remove every marker.

A folder with no addons inside and no `manifest.json` of its own counts as an addon, so you can enable, move and delete it like any other. Converted packages that the simulator writes, and scenery object libraries shipped without a manifest, look like this.

4.4 Pinning the destination

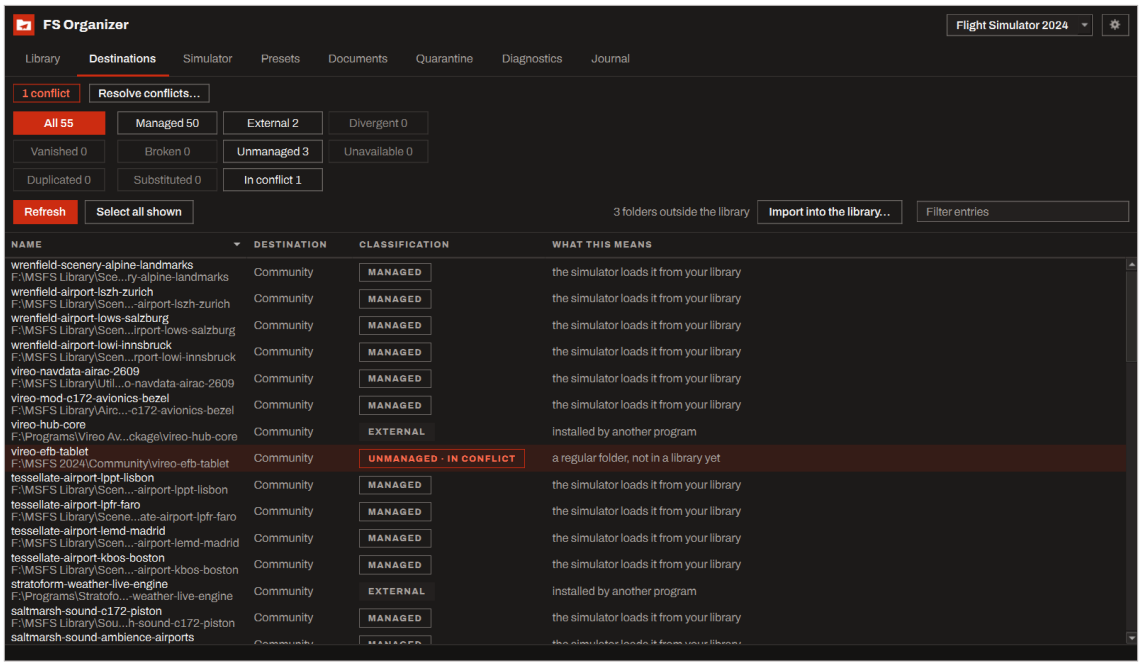
Addons are linked in the profile's default destination. **Always use** followed by a destination pins an addon, or a whole category, to that destination. If a pinned destination is removed from the profile, the program tells you and uses the default destination until you choose another. The pin itself is kept.

4.5 Relinking

If an addon is linked in the wrong destination, or its link points to a folder that no longer exists, **Relink in the profile destination** fixes it. Only the links change; the addon's files are not touched.

5 Destinations

The **Destinations** tab lists everything in the folders the simulator reads, including folders you did not add through FS Organizer. Each entry has a classification:



The destinations, with the classification filter at the top and the count of each kind at the bottom.

Managed
The simulator loads it from your library. This is the normal state.

External
A regular folder installed by another program, not in a library yet.

Divergent
The program that installed the addon put its own copy back, so there are two copies.

Vanished
The library copy is gone, removed by the other program. There is nothing to repair: the files are no longer on this computer.

Broken
The link points to a folder that does not exist.

Unavailable
The drive is not connected. It may come back, so the program does not offer to clean it up.

Unmanaged
A regular folder, not in a library yet.

Duplicated
Linked in more than one destination.

Substituted
Something wrote a regular folder over a link this program made. The simulator loads that folder, and enabling, presets and the culprit search no longer act on your library copy.

The filter at the top shows one classification at a time, and **Select all shown** selects exactly what is on screen.

5.1 Importing another program's folder

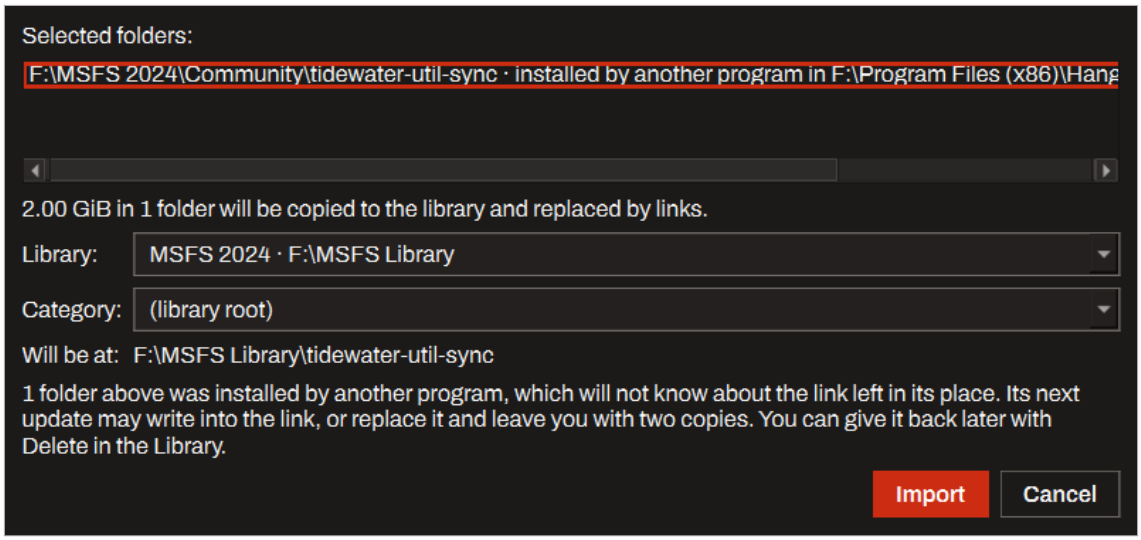
Import this folder moves the folder into your library and leaves a link in its place, so the addon becomes one of yours.

The other program will not know about the link. Its next update may write into the link, or replace it with a regular folder and leave you with two copies. When that happens, the entry shows as **Divergent** and the program asks which copy to keep.

To give the addon back to the program that installed it, use **Delete from the library**.

6 Importing

Importing copies a regular folder into the library and leaves a link where it was. The original is removed only after the copy has been verified. The journal records each of the five steps:



The import dialog: how much will be copied, into which library and category, and where the addon will end up.

1. Copy to a temporary folder.
2. Verify the copy.
3. Move the copy into place.
4. Remove the original folder.
5. Create the link in the destination.

While the copy runs, a window shows the current folder, the progress and a **Cancel** button. Cancelling deletes nothing: what was already copied is kept, so the import can be resumed later.

6.1 Verify after copying

On the **Links** tab in **Options**, you choose how the copy is verified.

By structure
Compares the number and size of the files. This is the default and it is fast.

By hash
Compares the content of every file. It catches more, but imports become several times slower, especially with many small files.

If the copy does not match, the original is not removed.

6.2 Unfinished imports

If an import is interrupted, the program offers three choices the next time it opens: resume the import, discard the partial copy, or leave it as it is. The original folder is untouched in every case.

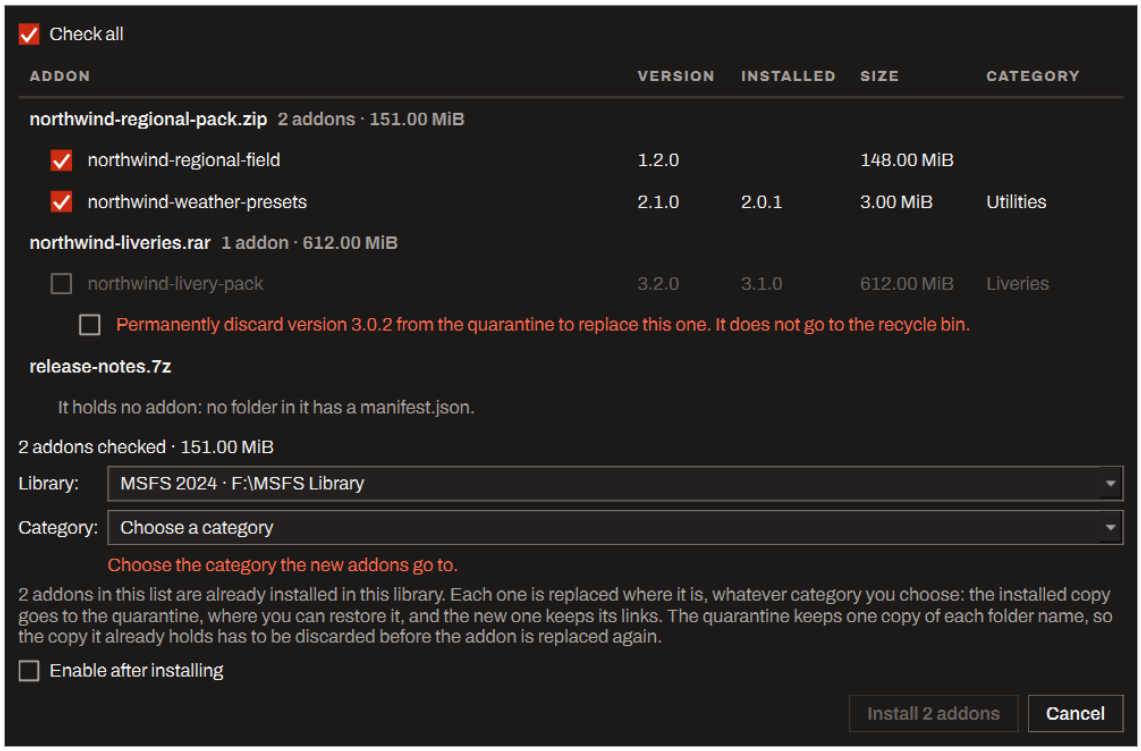
If the interrupted operation was a conflict resolution, the partial copy can only be discarded, because both copies are still in place.

7 Installing from an archive or a folder

Drop a `.zip`, `.rar` or `.7z` file, or a folder, onto the program's window. The program reads it without extracting anything and opens **Install into the library**, which lists every addon it found: each folder that has a `manifest.json`, with its version and size.

Choose the library and the category the new addons go to. Dropping onto a category in the **Library** tab chooses that category for you. Tick **Enable after installing** to have the addons enabled as soon as they are in the library.

An addon that is already in the library is replaced where it is, whatever category you choose, and the copy it replaces goes to the quarantine. The quarantine keeps one copy of each folder name, so when it already holds one, the dialog asks you to agree to discard it first.



The install dialog: two addons from one archive, one of them already installed, and a third that waits for the copy in the quarantine to be discarded.

The file or folder you dropped is never changed, moved or deleted. A folder is copied, an archive is extracted, and in both cases the result is checked against what was read before any addon is put into place.

7.1 What is refused

The program installs nothing from an archive or folder that:

- needs a password;
- holds no addon, or only an installer (`.exe` or `.msi`), which the program never runs;
- holds a link, an absolute path, `..` or a name Windows cannot write;
- holds two addons with the same folder name.

Nothing can be installed while the simulator is running. Close it and press **Check again** in the dialog.

7.2 What each Windows opens

Archives are opened by the `tar.exe` that comes with Windows, so what the program can open depends on your Windows version. A `.zip` opens on any of them. A `.rar` or `.7z` opens only if that Windows reads the format: Windows 10 does not read `.7z` or older `.rar` files. When an archive cannot be opened, extract it yourself and drop the folder instead.

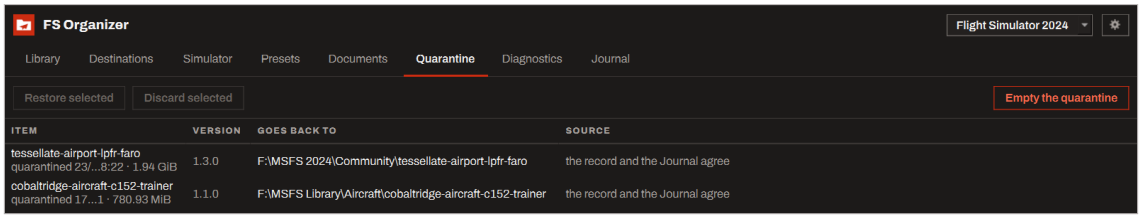
7.3 If an installation is interrupted

The files are extracted or copied into a temporary folder inside the category. If the program could not delete that folder, it appears under **Unfinished imports** the next time the program opens, where it can only be discarded.

8 Quarantine

When you resolve a conflict between two copies of an addon, the copy you do not keep goes to the quarantine instead of being deleted. Items stay there until you restore or discard them.

For each item, the program records where it came from, in a file next to the item. That record does not depend on the journal, so you can restore an item even if the journal is lost. The table shows both sources, **The record says** and **The Journal says**, and flags any disagreement.



The quarantine, with each item's origin and a flag where the record and the journal disagree.

Restore puts each folder back where it came from. If something is already there, the dialog shows both versions and you decide; replacing moves the current folder to the quarantine. **Discard** deletes the item permanently, after asking.

9 Deleting an addon from the library

Delete offers three ways to remove an addon, and tells you in advance when one of them will not work.

Move to the Recycle Bin
You can restore the addon from the Recycle Bin. This option is refused when the addon does not fit in that drive's Recycle Bin, or when it has a path longer than the 260 characters the Recycle Bin accepts. Windows may remove older items from the Recycle Bin to make room.

Delete permanently
This cannot be undone.

Give it back to the program that installed it
Only offered for an addon imported from another program's folder. The files go back to that folder and leave the library.

If the addon is enabled, it is disabled first.

10 Simulator

The **Simulator** tab has two panels, and both change files that belong to the simulator. Before changing one, the program saves a backup copy next to it. Neither panel changes anything while the simulator is running.

The startup entries start managed, and the package list starts unmanaged. While a panel is unmanaged, the program does not read or write its file. **Manage startup entries** and **Manage the package list** turn each one on, and the stop managing button on the panel's bar turns it off.

10.1 Startup entries

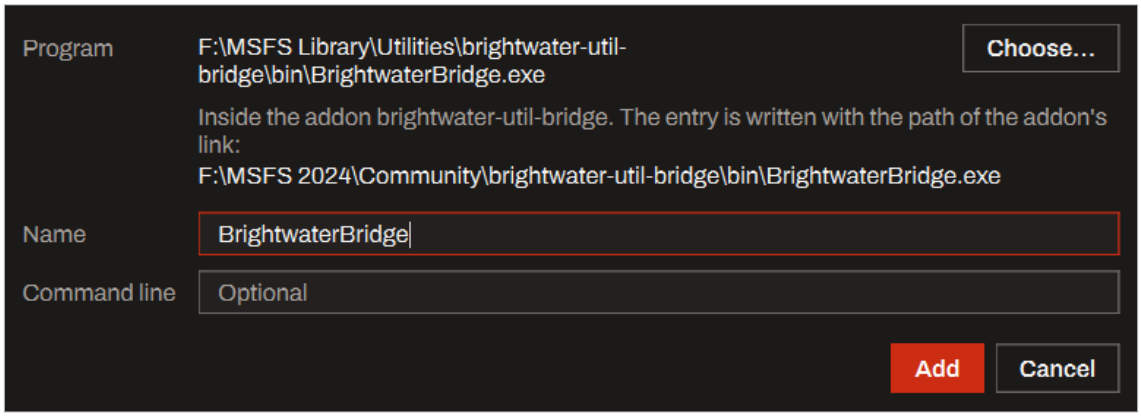
The simulator's `EXE.xml` file, next to its `UserCfg.opt`, lists the programs that start along with it. The **Startup entries** panel shows them and lets you enable, disable, add, edit and remove each one without editing XML. The program changes only the entry you asked for. The rest of the file stays as it was, and the other entries keep their place.



The startup entries, with the program each one launches, its switch and its state.

The **State** column says whether the entry's program is inside an addon of yours or outside your addons. When the program no longer exists, the entry gets the **the program is missing** tag. That is what an uninstalled program leaves behind, and it is usually the entry worth removing. If the program is in a disabled addon or on a drive that is not connected, the state says so instead of the tag, because in those two cases it can come back.

Add asks for the program, a name and, if you want, the command line. The entry starts enabled. Adding does not copy or move the program. If the profile has no `EXE.xml` yet, the file is created next to `UserCfg.opt`.



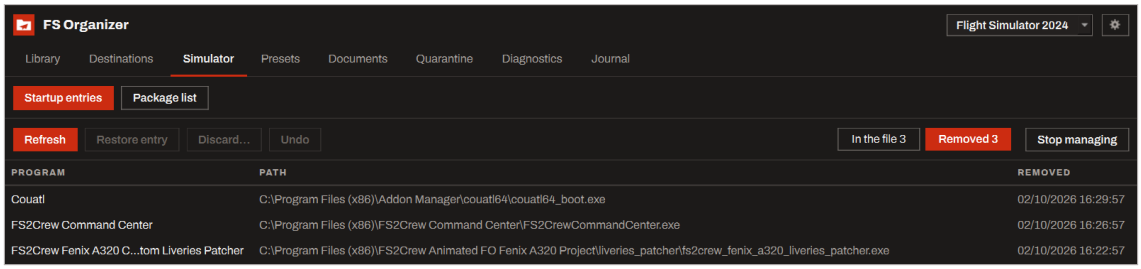
Adding a program that is inside an addon of the library. The entry is written with the path of the addon's link.

When the program you choose is inside an addon of your library, the entry is written with the path of the addon's link, which is what installers write. The window shows that path before you confirm. That way the entry follows the addon: disabling the addon warns you about it, as described further down.

Edit, a double click on the row or the F2 key open the same window with the selected entry. If you change the program, the presets that name the entry start naming the new path.

Remove, or the Delete key, takes the entry out of the file. Disabling already stops the simulator from launching the program, so removing is for the entry you no longer want to see in the list.

Every removed entry is kept under **Removed**, with its name, path and command line. **Restore entry** puts it back in the file as it was. **Discard** deletes it for good, and that cannot be undone. An edit that changes the program also keeps the entry there as it was before.



The removed entries, which can be restored or discarded.

The **Undo** button, or Ctrl+Z, takes back the last add, edit or removal made in this panel. It works while the program stays open. After that, the way back from a removal is the **Removed** list.

When you disable an addon that has a startup entry, the program warns you: otherwise the simulator would keep trying to launch a program that is no longer there. You can disable both in one step, which you can undo in one go.

10.2 Package list

The simulator ships its own airports. When one of your airports covers the same place as one of them, the **Package list** panel shows the pair and lets you disable the simulator's airport.

This changes one value in the simulator's package list. Nothing is added, removed or reordered. An entry that does not have that value is left alone, because an invalid file makes the simulator reset the whole list.

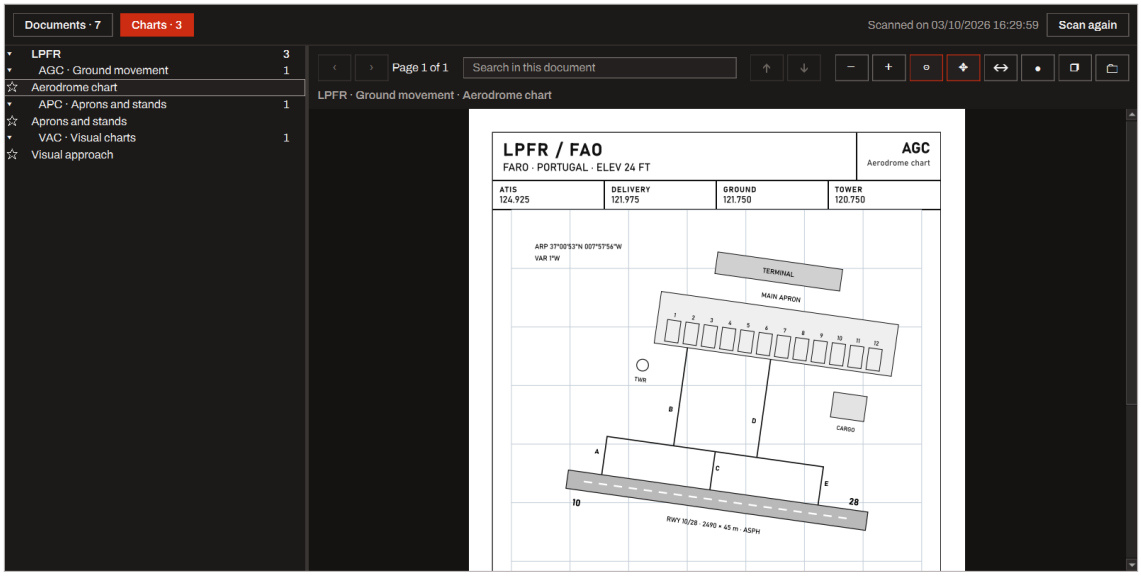
When two of your own addons cover the same airport, the program only shows the pair. To keep just one, disable the other on the **Library** tab.

11 Documents

The **Documents** tab finds the PDFs that come with your addons and opens them in the program. **Scan the library** looks for PDFs in every addon. The result is saved, so later scans are quick.

PDFs are split into **Documents** and **Charts**. Charts are grouped by airport, and also by type when the addon includes a catalogue of them. One addon can bring a manual and a hundred and fifty charts, so each group shows its count first.

The reader has an outline, search, fit to width and bookmarks. A bookmark you do not name is named after its page. **Open in a separate window** moves the reader out of the tab.

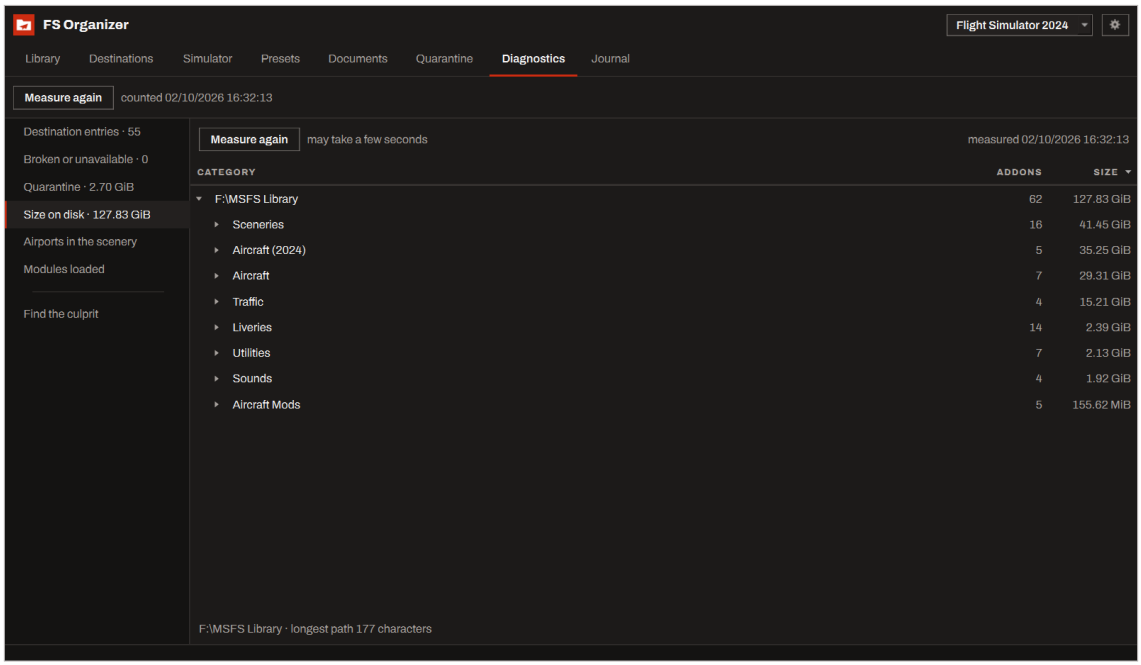


The documents tab: charts grouped by airport on the left, the reader in the middle and the document outline on the right.

On a chart, the mouse wheel zooms and dragging moves the page. On a document, both start off. The two buttons on the toolbar switch them.

12 Diagnostics

The **Diagnostics** tab shows what the program found on disk. Each section shows when it was measured, and you can measure it again at any time.



The size on disk section, with the total per category and per addon, and when it was measured.

Destination entries
How many entries there are of each kind, and the longest path in each library.

Broken or unavailable
Broken links and entries on a disconnected drive, with a button to repair the broken links.

Quarantine
How much space the quarantine uses.

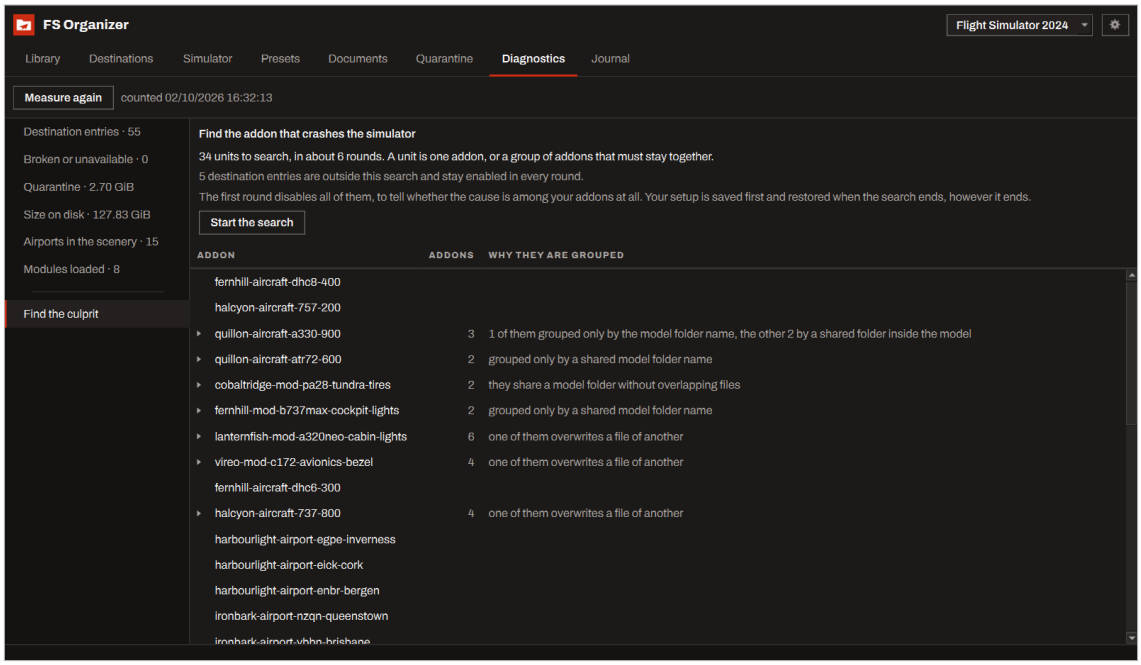
Size on disk
How much each category and addon takes up. It may take a few seconds; **Stop** stops after the current addon.

Airports in the scenery
Scans each addon's scenery and groups addons by what it finds: an airport code, an airport record that cannot be read, navigation data instead of scenery, or no airport record.

Modules loaded
Reads the report the simulator writes after a slow load. The simulator does not report loading time per package, so this shows which module each package loaded and how much memory it uses.

12.1 Finding the culprit

When the simulator crashes and you do not know which addon is responsible, **Find the culprit** narrows it down by halves. In each round, the program enables part of your addons, you launch the simulator and answer **It crashed** or **It worked**, and half of the remaining suspects are ruled out.



The culprit search, with the current round, what has been ruled out, and the two answers.

Before you start:

- The search works on units. A unit is one addon, or a group of addons that must stay together, for example because they share a model folder. The panel says why a group is together, and offers to split it when that is safe.
- The first round disables all your addons, to find out whether the cause is among them at all.
- Your setup is saved before the first round and restored when the search ends, however it ends. If the program closes during the search, it offers to restore your setup the next time it opens.
- The search assumes a single culprit. If the crash only happens when two addons are enabled together, the result may point to the wrong one.

13 Journal

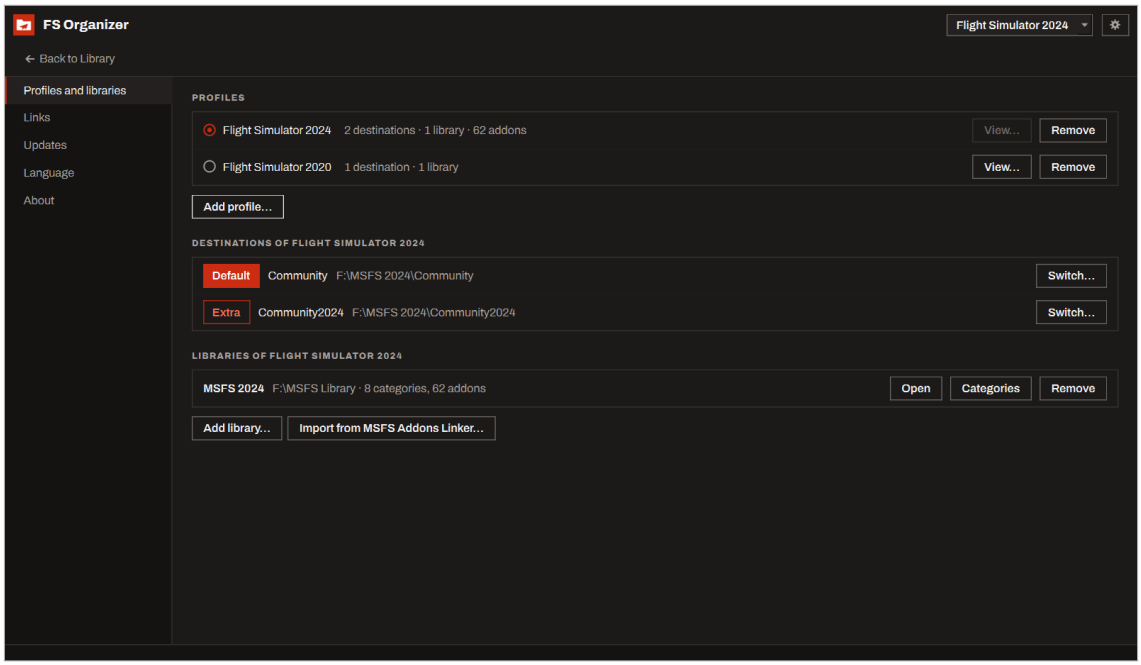
Every operation that changes files on disk adds a line to the **Journal**. Lines are only ever added, and the program never deletes the journal.

Each line shows when it happened, the operation, the addon, the library, the source, the destination and the result. **Failures only** shows just what went wrong, and the search box looks in addon names, paths and operations.

Undo last change, on the **Library** tab, undoes the whole last change, not a single line of it.

14 Options

The gear button in the header opens the options. They have five tabs.



The profiles and libraries tab.

Profiles and libraries
Choose the profile in use, add or remove profiles and libraries, and change destinations. Removing a library deletes no files: its links keep working in the simulator, but FS Organizer stops managing them. **Categories** shows how many category folders are marked and lets you mark them or remove the markers.

Links
The link type, and how imports are verified.

Updates
Automatic, notify, or manual. Automatic downloads new versions and installs them when you close the program.

Language
English or Brazilian Portuguese. The change is immediate.

About
The version, the licence and the credits.

15 Where things are kept

The program keeps its data in `%LOCALAPPDATA%\fs-organizer`. `settings.json` holds the profiles, libraries, destinations and your choices. `journal\operations.jsonl` is the journal. `presets` holds one file per preset, plus each profile's return preset. `bisection` holds the state of a culprit search in progress, and is deleted when the search ends.

None of your addons are stored there. If you delete that folder, you lose the configuration and the presets, but no addon: the program finds them again by reading the destinations and libraries.

The only thing the program writes outside that folder is the hidden `.fsorg-category` marker in your category folders. **Categories** in the options removes all of them.